



Methane and its derivative			
family	Name	Mol. formula	Structural Formula
Hydro Carbon	Methane	$\text{CH}_4^{\checkmark}$	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$
② Saturated (Alkane)	Ethane	$\text{C}_2\text{H}_6^{\checkmark}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$
General formula $\text{C}_n\text{H}_{2n+2}$	Propane	C_3H_8	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$
[Paraffin] (less friendship)	But-ane	C_4H_{10}	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array}$
③ Unsaturated hydro Carbon (double bond alkene)	Ethylene	$\text{C}_2\text{H}_4^{\checkmark}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{H} \end{array}$
Triple bond	Acetylene	$\text{C}_2\text{H}_2^{\checkmark}$	$\text{H}-\text{C} \equiv \text{C}-\text{H}$
Alcohol —OH	Methyl Alcohol (Methanol)	$\text{CH}_3\text{OH}^{\checkmark}$	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{H} \end{array}$



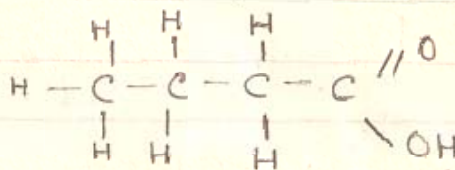
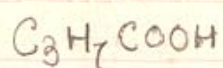
Ethyl Alcohol (Ethanol)	C_2H_5OH	$ \begin{array}{c} H & H \\ & \\ H - C & - C - OH \\ & \\ H & H \end{array} $
Propyl Alcohol (Propanol)	C_3H_7OH	$ \begin{array}{c} H & H & H \\ & & \\ H - C & - C - C - OH \\ & & \\ H & H & H \end{array} $
Butyl Alcohol (Butanol)	C_4H_9OH	$ \begin{array}{c} H & H & H & H \\ & & & \\ H - C & - C - C - C - OH \\ & & & \\ H & H & H & H \end{array} $
Pentanol	$C_5H_{11}OH$	$ \begin{array}{c} H & H & H & H & H \\ & & & & \\ H - C & - C - C - C - C - OH \\ & & & & \\ H & H & H & H & H \end{array} $
Glycol	$ \begin{array}{c} CH_2OH \\ \\ CH_2OH \end{array} $	$ \begin{array}{c} H \\ \\ H - C - OH \\ \\ H - C - OH \\ \\ H \end{array} $
Glycerol or Glycerine	$ \begin{array}{c} CH_2OH \\ \\ CH_2OH \\ \\ CH_2OH \end{array} $	$ \begin{array}{c} H \\ \\ H - C - OH \\ \\ H - C - OH \\ \\ H - C - OH \\ \\ H \end{array} $
Halogen derivative	Chloroform (Trichloro Methane)	$ \begin{array}{c} Cl \\ \\ Cl - C - H \\ \\ Cl \end{array} $
Iodoform	CHI_3	$ \begin{array}{c} I \\ \\ I - C - H \\ \\ I \end{array} $



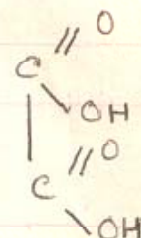
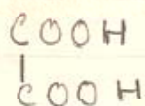
Bromoform (TriBromo- Methane)	CHBr_3	$\begin{array}{c} \text{Br} \\ \\ \text{Br} - \text{C} - \text{H} \\ \\ \text{Br} \end{array}$
Ethyl Chloride	$\text{C}_2\text{H}_5\text{Cl}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H} - \text{C} - \text{C} - \text{Cl} \\ \quad \\ \text{H} \quad \text{H} \end{array}$
Ethyl Bromide	$\text{C}_2\text{H}_5\text{Br}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H} - \text{C} - \text{C} - \text{Br} \\ \quad \\ \text{H} \quad \text{H} \end{array}$
Ethyl Iodide	$\text{C}_2\text{H}_5\text{I}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H} - \text{C} - \text{C} - \text{I} \\ \quad \\ \text{H} \quad \text{H} \end{array}$
Propyl Iodide	$\text{C}_3\text{H}_7\text{I}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H} - \text{C} - \text{C} - \text{C} - \text{I} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$
Methyl Bromide	CH_3Br	$\begin{array}{c} \text{H} \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{H} \end{array}$
Acids -COOH (Carboxylic group)		
Formic Acid (Methanoic Acid)	HCOOH	$\begin{array}{c} \text{O} \\ \\ \text{H} - \text{C} \\ \backslash \\ \text{OH} \end{array}$
Acetic Acid (Ethanoic Acid)	CH_3COOH	$\begin{array}{c} \text{H} \quad \text{O} \\ \quad \\ \text{H} - \text{C} - \text{C} \\ \quad \backslash \\ \text{H} \quad \text{OH} \end{array}$
Propionic Acid	$\text{C}_2\text{H}_5\text{COOH}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \\ \text{H} - \text{C} - \text{C} - \text{C} \\ \quad \quad \backslash \\ \text{H} \quad \text{H} \quad \text{OH} \end{array}$



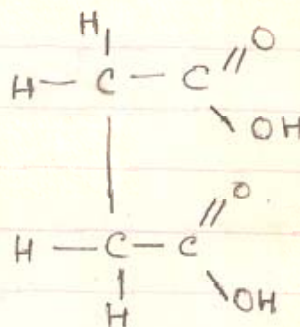
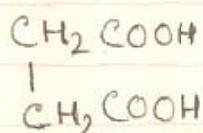
Butanoic
Acid



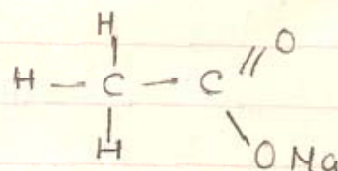
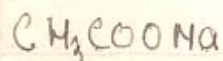
Oxalic
Acid



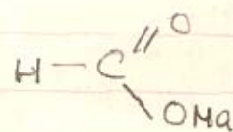
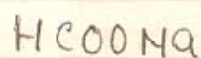
Succinic
Acid



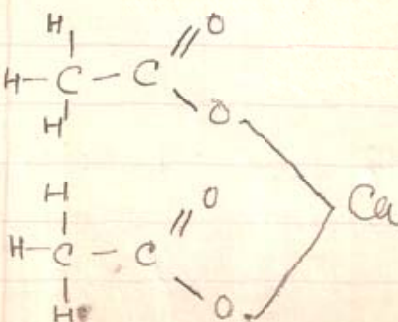
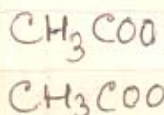
Sodium
Acetate



Sodium
Formate

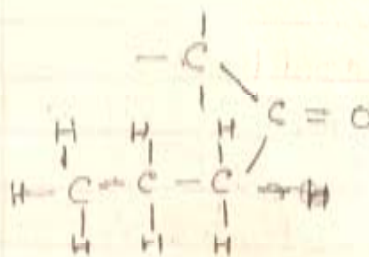
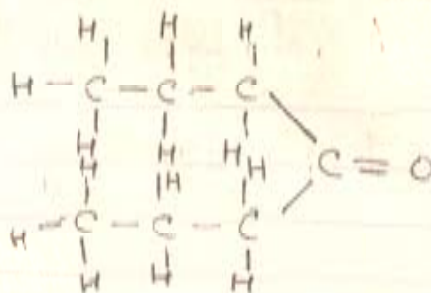
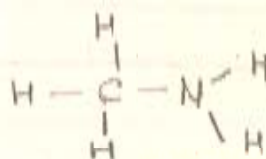
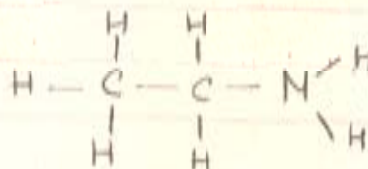
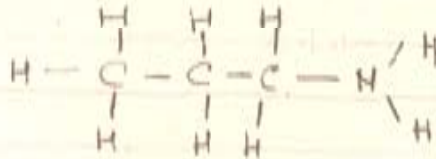


Calcium
Acetate

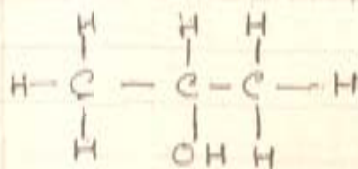
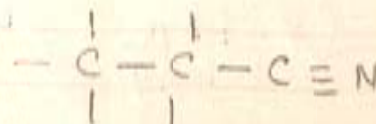


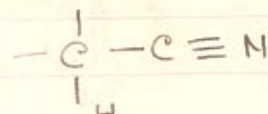
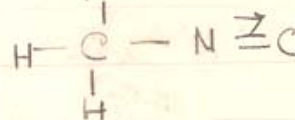
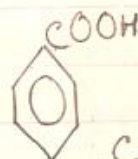
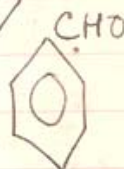
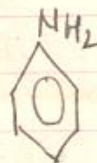
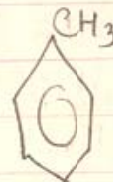
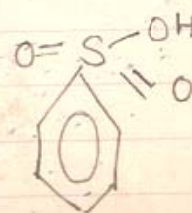


Aldehyde -CHO $\begin{array}{c} \text{O} \\ \parallel \\ \text{C} \\ \diagup \quad \diagdown \\ \text{H} \end{array}$	Formaldehyde (Methanol) 40% of it is called formalin	HCHO ✓	$\text{H}-\text{C} \begin{array}{l} \parallel \text{O} \\ \diagdown \text{H} \end{array}$
	Acetaldehyde	CH_3CHO ✓	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{C} \begin{array}{l} \parallel \text{O} \\ \diagdown \text{H} \end{array} \\ \\ \text{H} \end{array}$
	Propionaldehyde	$\text{C}_2\text{H}_5\text{CHO}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{C} \begin{array}{l} \parallel \text{O} \\ \diagdown \text{H} \end{array} \\ \quad \\ \text{H} \quad \text{H} \end{array}$
	Butanaldehyde	$\text{C}_3\text{H}_7\text{CHO}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \begin{array}{l} \parallel \text{O} \\ \diagdown \text{H} \end{array} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$
Ether RO-R'	Common Ether (Diethyl Ether)	$\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ ✓	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array}$
	Methyl Propyl Ether	$\text{CH}_3\text{OC}_3\text{H}_7$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{O}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$
	Di Methyl Ether	CH_3OCH_3	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{O}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$
Ketone $\begin{array}{c} \text{R} \quad \text{R}' \\ \diagdown \quad \diagup \\ \text{C}=\text{O} \end{array}$	Acetone (Di Methyl Keton)	CH_3COCH_3 ✓	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}=\text{O} \\ \quad \\ \text{H} \quad \text{H} \end{array}$

$$\text{CH}_3\text{COC}_3\text{H}_7$$

$$\text{C}_3\text{H}_7\text{COC}_3\text{H}_7$$

$$-\text{NH}_2$$
$$\begin{array}{c} \text{H} \\ | \\ \text{H}-\text{C}-\text{H} \\ | \\ \text{H} \end{array}$$
 CH_3NH_2 
$$\text{C}_2\text{H}_5\text{NH}_2$$

$$C_3H_7NH_2$$


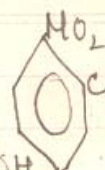
Iso Propyl Alcohol


$$\text{C}_2\text{H}_5\text{CN}$$


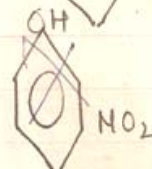
Methyl Cyanide CH_3CN Methyl Isocyanide CH_3NC Benzene C_6H_6  C_6H_5 - PhenylNitro Benzene $\text{C}_6\text{H}_5\text{NO}_2$ Benzoic Acid $\text{C}_6\text{H}_5\text{COOH}$ Benzaldehyde $\text{C}_6\text{H}_5\text{CHO}$ Phenol
(Carbolic Acid) $\text{C}_6\text{H}_5\text{OH}$ Aniline / Phenyl Amine $\text{C}_6\text{H}_5\text{NH}_2$ Toluene / Methyl Benzene $\text{C}_6\text{H}_5\text{CH}_3$ Chloro Benzene $\text{C}_6\text{H}_5\text{Cl}$ Benzene Sulphonic Acid $\text{C}_6\text{H}_5\text{SO}_3\text{H}$ 



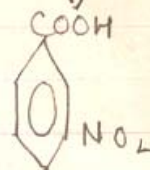
Ortho chloro Nitro Benzene



Meta Nitro Benzoic Acid



Meta Nitro Benzoic Acid



Picric Acid (present in Burrenol) / 2:4:6 Trinitrophenol

